

A Primary counterfactual

Primary measurement
PortWatch Hormuz tanker transits
1 Jan 2022–7 Jul 2026

Leakage-controlled forecast
train strictly before 28 Feb 2026
recursive AR(1,7) • 130-day recursive path
daily and cumulative shortfall: $\hat{y}_t^0 - y_t$

Chronological validation and uncertainty
30-day rolling-origin folds
+ 8 disjoint 130-day reference blocks
rank and conformal calibration

Falsification and corroboration
temporal and spatial placebos
synthetic control • model and window checks
capacity and vintage sensitivity

B Descriptive adaptation

Supporting measurements
WTO index • GFW events and route model
corridor counts • customs portfolios
source-specific units and windows

Within-layer adaptation metrics
LNG activity • inferred terminal sequences
nominal capacity-distance
corridor deviations
importer-origin portfolios

Comparable calendars, separate scales
matched windows within each source
no pooling of non-equivalent units

Triangulation by direction and timing
contraction, substitution, and reallocation
disagreement retained as evidence

Permitted synthesis
Disruption-associated counterfactual shortfall in observable tanker throughput
+ separately bounded evidence on LNG and shipping-network adaptation
Not a causal effect, physical census, freight-rate estimate, or identified cargo replacement